AFAG RAMAZANOVA
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Education

Duke University North Carolina, US | Expected May 2026
*Master of Science in **Data Science**; GPA:3.9 | Full Scholarship Recipient | Teaching Assistant*
Coursework: Machine Learning, Data Manipulation, Statistical Modeling, Data Visualization, Data Engineering

Baku Engineering University Baku, Azerbaijan | Sep 2016 – June 2021
Bachelor's degree in Economics: GPA:3.8 | Full Scholarship Recipient
Summer School participant @ Koblenz University of Applied Sciences

Experience

AI for Compliance Capstone Researcher | *BNY Mellon* Remote | Aug 2025 – present

- Designed an AI-driven compliance report generation system using RAG and LLMs to automate AML regulatory narratives.
- Built a regulatory knowledge base from financial regulations (OFAC, FINCEN, FINRA, AML/KYC) enabling citation-based retrieval for compliance analysis.
- Developed a multi-agent architecture for ingestion, contradiction detection, and automated report drafting with human-in-the-loop validation.

Data Science Intern | *Citizens Bank* Columbus, Ohio | May 2025 – Aug 2025

- Developed a predictive model for Smart Leads to **estimate customer contactability** using historical engagement data, enabling prioritization of high contactability leads and reducing outreach inefficiencies across key business units.
- Performed end-to-end analysis on upper funnel marketing investment and their impact on brand awareness using Adstock and S-curve transformations, providing strategic insights for media allocation.
- Built a regression model to measure how marketing investments impact brand awareness, guiding media investment across key markets.
- Designed an automated **simulation tool** to forecast brand awareness based on hypothetical investment scenarios, supporting future marketing planning.

Data Scientist | *Kapital Bank* Baku, Azerbaijan | Aug 2022 – Aug 2024

- Developed and maintained predictive models including Probability of Default, Income Estimation, Debt collection models to identify customers' risk.
- Initiated the deployment of model as an API service using Dataiku, reducing the implementation time from **one month to 3 days.**
- Introduced a new predictive model that led to a **25%** increase in the bank's retail credit portfolio and a significant rise in sales.
- Streamlined reporting processes by designing automated dashboards, enabling daily performance tracking of new model, and improving response times to problems by **30%**, significantly enhancing stakeholder engagement.

Data Scientist | *Unibank* Baku, Azerbaijan | May 2021 – Aug 2022

- Developed an automated reporting system that reduced manual work by 50% and improved response times for risk inquiries by 20%.
- Used reports and insights to monitor the performance of existing loans and lines of credit leading to **10%** reduction in the bank's nonperforming loan ratio through improved credit risk assessment processes.
- Facilitated the development of a model risk strategy by providing insights into the importance of model risk management and its benefits, helping the bank address an area previously not covered.

Skills

Programming Languages, Platforms & Tools: Python, SQL, AWS, GitHub, R Studio, Tableau, MS Office Suite, SPSS Statistics,Claude Code, Cursor, n8n

Data Manipulation Visualization: Pandas, NumPy, Matplotlib, Seaborn, ggplot2, dplyr, tidyr

Functional Expertise: LLMs, Agentic Workflows, Prompt engineering Regression, Forecasting, Data Mining, Feature Engineering, Agile, A/B Testing, Web Scraping, CI/CD

Soft skills: Problem-solving, critical thinking, communication, team collaboration, time management and prioritization

Certificates: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, Dataiku ML Practitioner

Projects

AskSQL: Project link Dec 2024
Tech: *AWS RDS, Python, AWS Bedrock, AWS AppRunner*

- Built a microservice that turns natural language questions into SQL queries, making it easier to interact with databases.

Customer Default Prediction Model: Project link May 2024
Tech: *SQL, Python, Dataiku, MS Excel, EDA*

- Developed a probability of default (PD) model to assess customer credit risk, achieving a GINI coefficient of 65%. **Term**

Deposit Subscription Prediction: Project link May 2023
Tech: *Python*

- Built a machine learning model to predict customer subscription rates for term deposits based on historical data. **Fashion**

Product Reviews Sentiment Analysis: Project link June 2022
Tech: *Scikit-learn, Pandas, Matplotlib, Seaborn, EDA*

- Conducted sentiment analysis of fashion product reviews, identifying key features and extracting customer opinions from text data.